11.24 Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

a. Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for ln(*PRICE*), adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?

b. Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.

c. Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.

d. In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

11.24

a. Reduced-form for ln(PRICE):

$$ln(Price) = -0.3595685 + mon * -0.1082483 + tue * -0.06609207 + wed * -0.04927356 + thu * 0.03934669 + stormy * 0.4463387 + mixed * 0.2368753$$

Summary statistic:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.35957    0.07915  -4.543 1.50e-05 ***
mon         -0.10825    0.10339  -1.047  0.29753    
tue         -0.06609    0.10103  -0.654  0.51446    
wed         -0.04927    0.10377  -0.475  0.63591    
thu          0.03935    0.10071   0.391  0.69681    
stormy       0.44634    0.07921   5.635 1.51e-07 ***
mixed        0.23688    0.07851   3.017  0.00321 ** 
```

Since p-value of both stormy and mixed are smaller than 0.05, which means that they are significant at 5% level.

```
Linear hypothesis test:
stormy = 0
mixed = 0

Model 1: restricted model
Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

  Res.Df    RSS Df Sum of Sq     F    Pr(>F)    
1    106 15.876                                 
2    104 12.115  2    3.7604 16.14 7.857e-07 ***
```

The joint F-statistic for STORMY and MIXED is 16.14, which is greater than 10, indicating that the instruments are strong and relevant.

b. Demand equation using stormy and mixed are IVs:

$$ln(Quantity) = 8.540234 + lprice * -0.9301409 + mon * -0.01190203 + tue * -0.04927356 + wed * -0.5626197 + thu * 0.09987059$$

Summary statistic:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.54023    0.15661  54.533  < 2e-16 ***
lprice      -0.93014    0.35340  -2.632  0.00977 **
mon         -0.01190    0.20837  -0.057  0.95456
tue         -0.52583    0.20230  -2.599  0.01068 *
wed         -0.56262    0.20696  -2.719  0.00767 **
thu          0.09987    0.20285   0.492  0.62350
```

Compare to Table 11.5, the effect of variables have smaller effect in (b), and the significance of variables are same as Table 11.5, and it is consistent with the law of demand.

c.

```
Diagnostic tests:
                 df1 df2 statistic  p-value
Weak instruments   2 104    16.140 7.86e-07 ***
Wu-Hausman         1 104     1.489    0.225
Sargan             1  NA     0.772    0.380
```

The p-value of Sargan test is 0.38, which means that we can conclude that all the IVs are valid and exogenous.

d.

The model satisfies the order condition because each equation excludes at least one exogenous variable. The instruments (STORMY and MIXED) are relevant as they are jointly significant in the reduced-form regression. The Sargan test fails to reject the null hypothesis, indicating that the instruments are exogenous. Therefore, the model is identified and can be estimated using 2SLS.

11.28

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

a. Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?

b. Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?

c. Estimate the price elasticity of demand "at the means" using the results from (b).

d. Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI, PS, and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.

e. What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?

f. Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

a. Demand: $P_i = a_1 + a_2Q_i + a_3PS_i + a_4DI_i + e_{di}$

Supply: $P_i = b_1 + b_2Q_i + b_3PF_i + e_{si}$

b.

Coefficient for demand function

```
             Estimate Std. Error  t value   Pr(>|t|)
(Intercept) -11.42841   13.59161 -0.84084  0.4081026
q            -2.67052    1.17495 -2.27287  0.0315350 *
ps            3.46108    1.11557  3.10252  0.0045822 **
di           13.38992    2.74671  4.87490 4.6752e-05 ***
```

In demand function, price and quantity should be negatively correlated, price and substitution price should be positively correlated (demand curve shift rightward), price and income should be positively correlated (demand curve shift rightward).

All of them are correct, and they are significantly different from 0 except intercept.

Coefficient for supply function

```
              Estimate Std. Error  t value   Pr(>|t|)
(Intercept) -58.798223   5.859161 -10.0353 1.3165e-10 ***
q             2.936711   0.215772  13.6103 1.3212e-13 ***
pf            2.958486   0.155964  18.9690 < 2.22e-16 ***
```

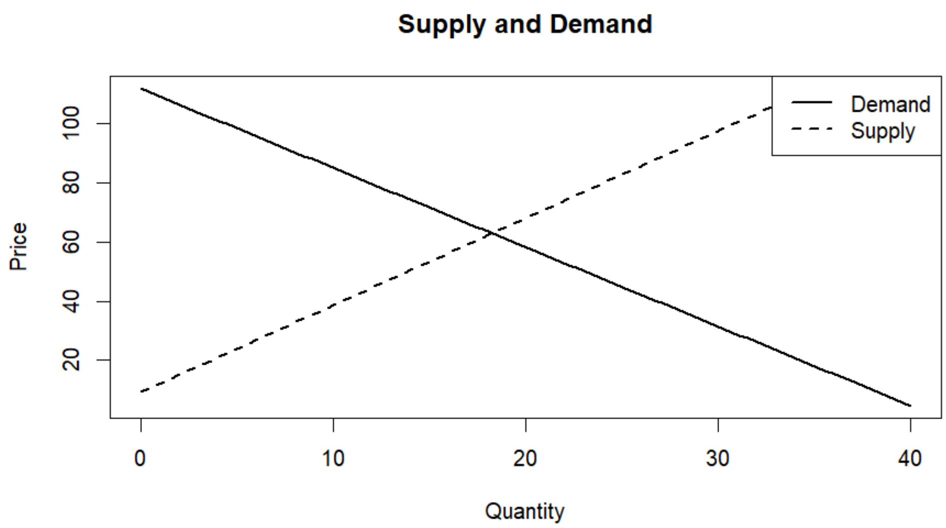
In supply function, price and quantity should be positively correlated, price and price factor production should be positively correlated (supply curve shift leftward).

All of them are correct, and they are significantly different from 0.

c. Elasticity of demand function is defined as: $\varepsilon = \frac{dQ}{dp} * \frac{\bar{P}}{\bar{Q}}$. Hence the elasticity is

$\frac{1}{-2.67052} * \frac{\bar{P}}{\bar{Q}} = -1.272464$

d. According to the coefficient in (b), we have



e. According to the model, we have $\begin{cases} Demand: P = a_2Q + c_1 \\ Supply: P = b_2Q + c_2 \end{cases}$, $where\ c_1 = a_1 + a_3PS_i + a_4DI_i, c_2 = b_1 + b_3PF_i,$ then $Q = \frac{c_2 - c_1}{a_2 - b_2}, P = \frac{a_2c_2 - b_2c_1}{a_2 - b_2}$

We have Equilibrium Quantity: 18.25021, Equilibrium Price: 62.84257, and with reduced-form, we have Reduced-form Quantity: 18.2604, Reduced-form Price: 62.81537.

The results are nearly identical, confirming that the reduced form is consistent with the structural model.

f.

(i) Coefficient for demand function

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -13.6195     9.0872  -1.499   0.1460    
q             0.1512     0.4988   0.303   0.7642    
ps            1.3607     0.5940   2.291   0.0303 *  
di           12.3582     1.8254   6.770 3.48e-07 ***
```

By economic intuition in (b),

The sign of coefficient for demand function quantity is incorrect and is not significant from 0.

The coefficients on substitution price and income remain positive and statistically significant, consistent with economic theory.

(ii)Coefficient for supply function

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -52.8763     5.0238  -10.53 4.68e-11 ***
q             2.6613     0.1712   15.54 5.42e-15 ***
pf            2.9217     0.1482   19.71  < 2e-16 ***
```

The sign of coefficient for supply function quantity is correct, and price factor production is correct. All of them are significantly different from 0.

Although the sign is correct, the magnitude differs from the 2SLS estimate, indicating bias.

OLS fails to recover the true structural relationships, while 2SLS provides consistent estimates by using instrumental variables to address endogeneity.